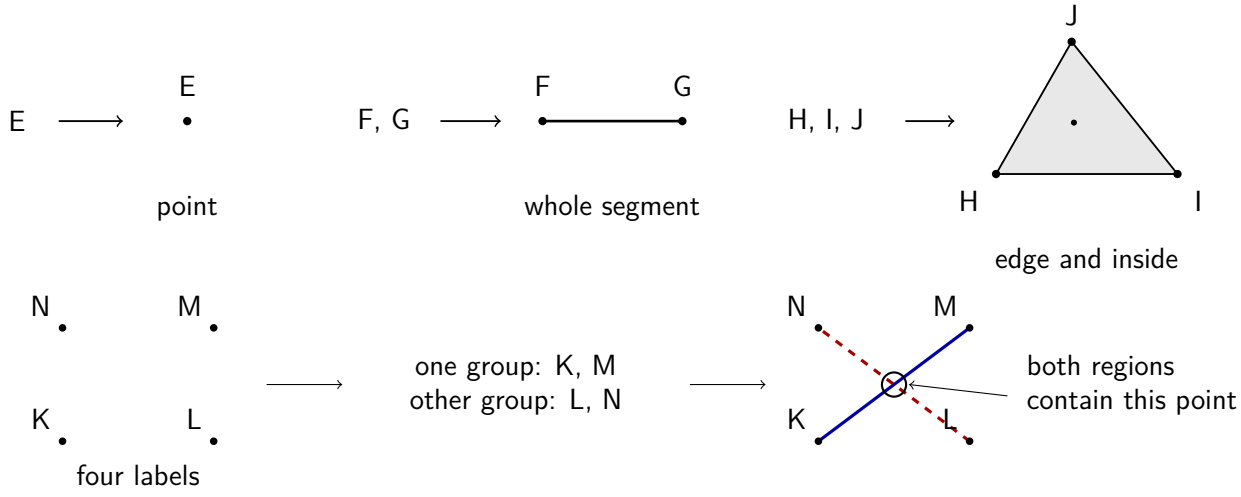
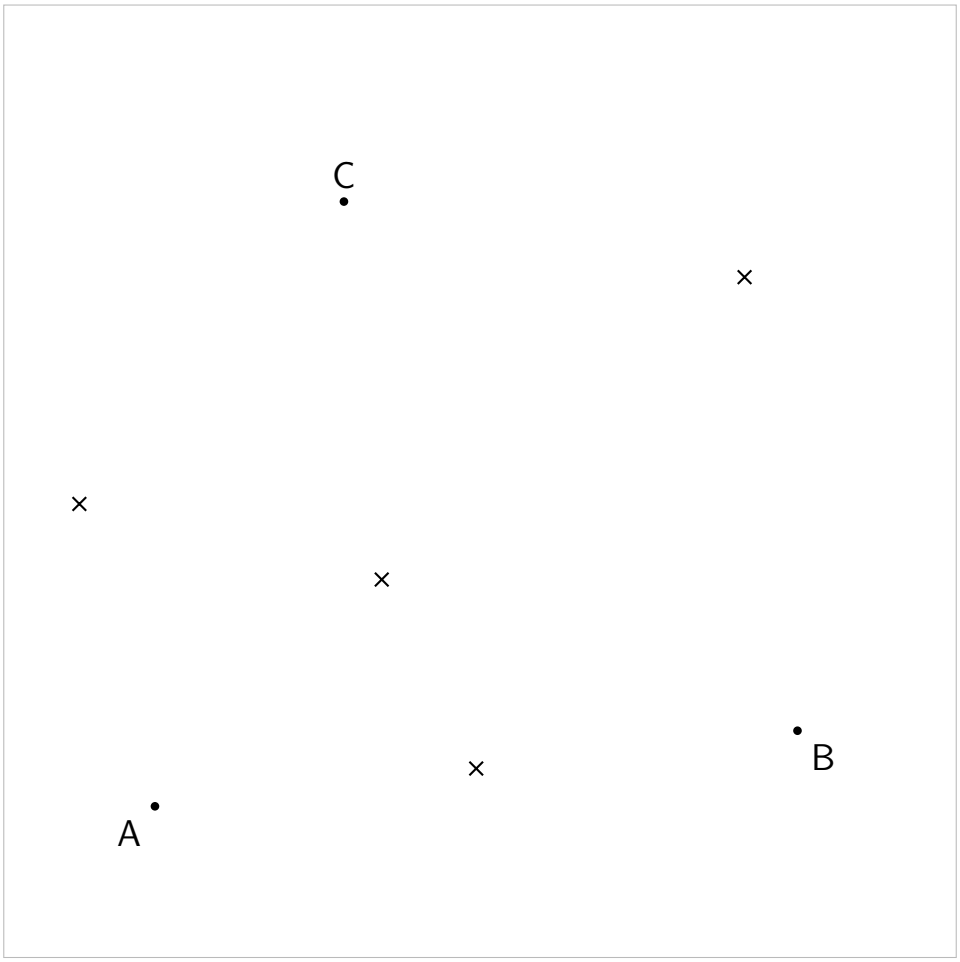
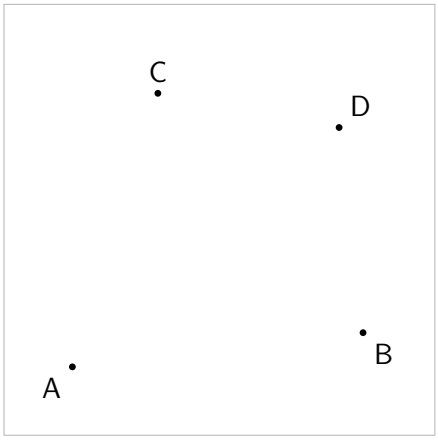
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split. When dots lie on one line, draw one group's region just above them and the other group's region just below.

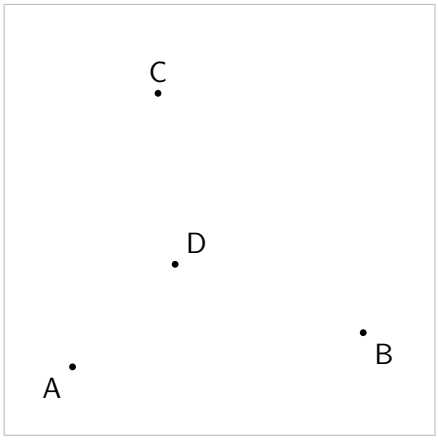


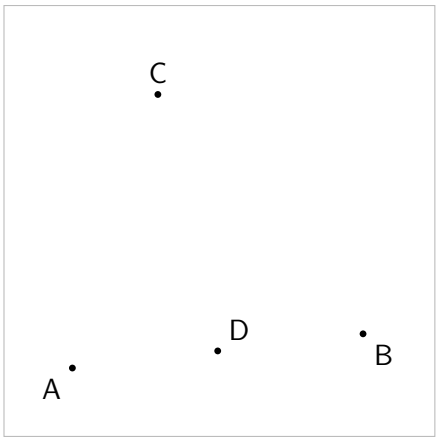
Problem 1: Move the D marker to each cross in turn. Each time, use the markers and your finger to find a split of A, B, C, and D whose two regions meet.

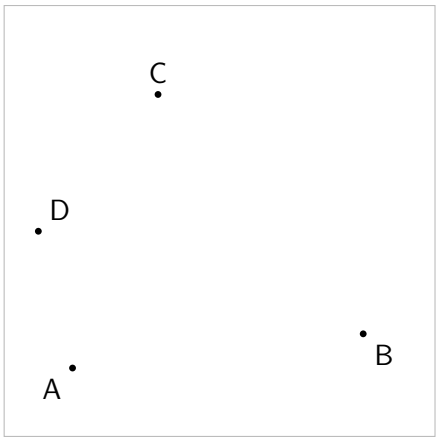


Problem 1: Draw the split you found for each position of D, and mark a point its two regions share.

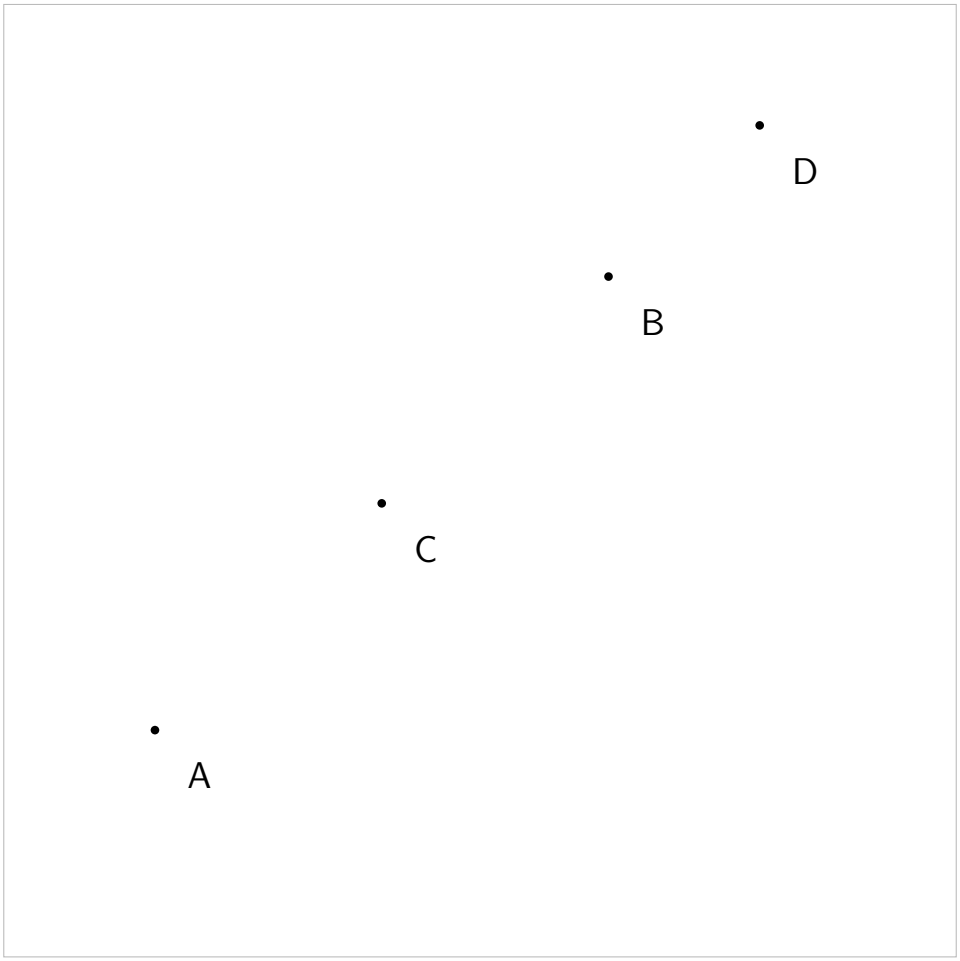




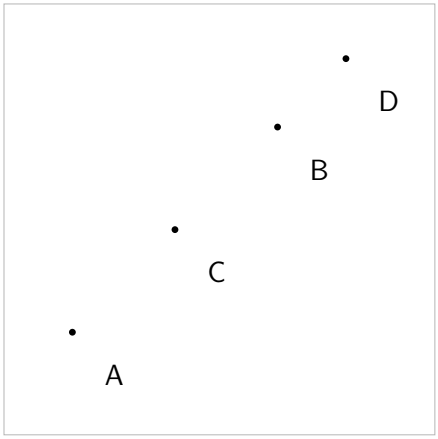


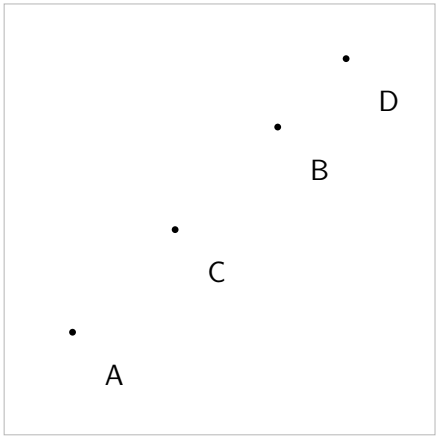


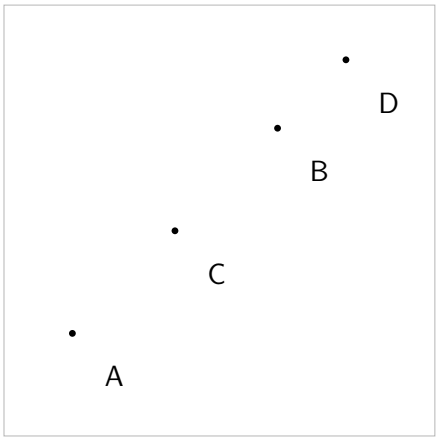
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

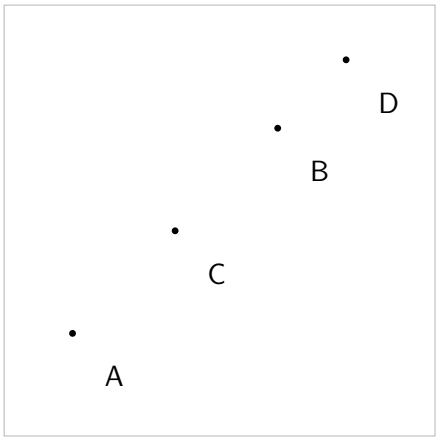


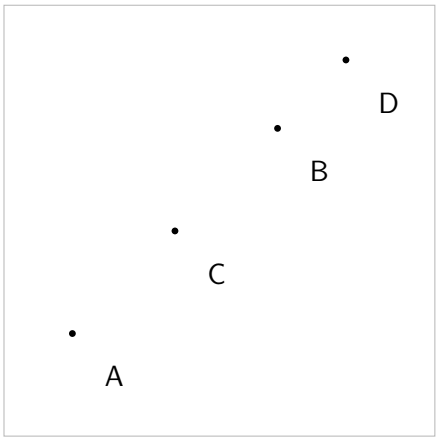
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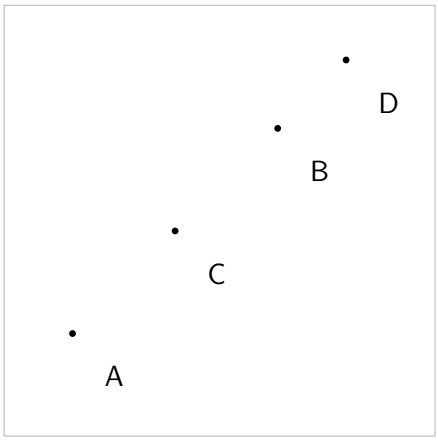












Problem 3: Take turns placing A, B, C, and D in the box. Try to make an arrangement with no successful split. Decide whether your opponent must always be able to find one.

Problem 4: A and D share a location. Find every split whose regions meet. Does a successful split always exist when two of four labels share a location? Explain.



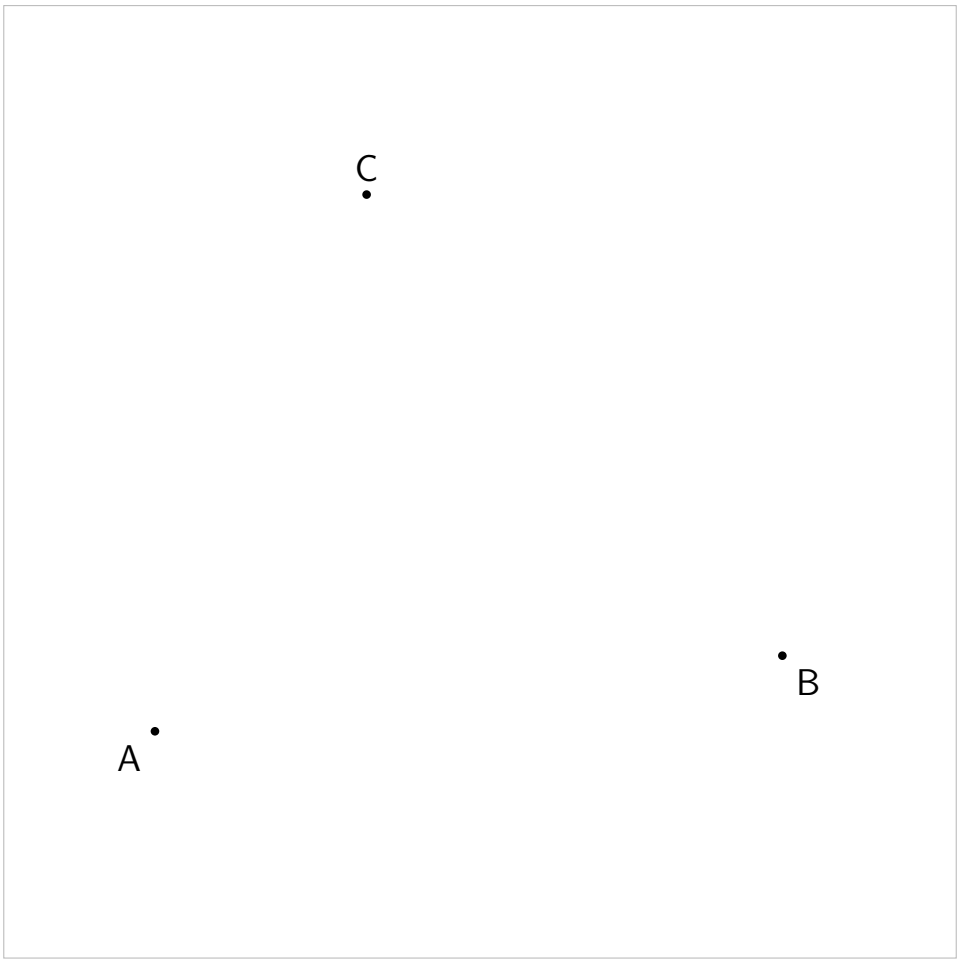
Problem 5: Make two four-dot arrangements that each have exactly one successful split, with different successful splits. Make a third arrangement with more than one successful split. Explain how you checked the number of successful splits.

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Problem 6: Can every arrangement of A, B, C, and D be split into two groups whose regions meet? Give an explanation that covers all arrangements.

Problem 7: Can these three dots be split into two groups whose regions meet? Explain why your answer covers every split.



Problem 8: Find the smallest number of dots that guarantees a successful split for every arrangement. Explain why fewer dots cannot give that guarantee.